Advanced Inorganic and Photochemistry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Inorganic and Photochemistry (A07604)

Short Title:	Adv Inorganic & Photochemistry		
Faculty	Science and Computing		
Department	Science		
Cluster	Chemistry		
Author	HHUGHES		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module gives the student and in-depth knowledge of bonding, formation and interpretation of organometallic complexes and catalytic cycles. The student should also be to apply point group theory to molecules to determine their structure. A detailed understanding of bioinorganic compounds of interest, including pharmaceutical products will be given. Students will also get an introduction to the significance of photochemical processes in a number of scientific fields including, synthetic chemistry, analytical chemistry and environmental science.

Programmes

CHEM-0034	BSc (Hons) in Applied Chemistry with Quality Management (WD_SQMCH_B)	4 / 2 / M
CHEM-0034	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	4 / 8 / M

Indicative Content

- Organometallic chemistry: chemistry of metal carbonyls and metal alkyls, methods of formation, bonding
- Catalytic cycles: properties of catalysts using transition metal complexes, examples of catalytic cycles such as acetic acid synthesis, Ziegler-Natta process, Wilkinson Catalyst
- Point group theory: assigning point groups, determining the number of IR active band in a spectrum of a molecule, use of point group theory
- Bioinorganic chemistry including Heme cycle, photosynthesis, anti-cancer drugs, As- & Fe-based antibiotics
- Photochemical deactivation processes, advantages/disadvantages of photochemical reactions
- Photophysical processes, fluorescence, phosphorescence, etc. (significance in pharmaceutical analysis and modern sensing)
- Photochemical photodissociation, photosensitised reactions etc
- Efficiency of photochemical reactions: quantum yield measurements, primary and overall QY; measurement of QY
- Applications of photochemistry: inorganic, solar energy conversion; atmospheric reactions; environmentally significant reactions
- Photochemistry and pharmaceutical stability

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Distinguish the bonding, methods of formation and analysis of orgnometallic complexes.
2. Assess the reaction mechanism for a number of fundamental catalytic cycles and account for their reactivity.
3. Formulate point groups and predicting infra-red spectra of complexes.
4. Assess the use of bioinorganic molecules in different applications.
5. Distinguish the properties and possible fates of photochemically excited molecules.
6. Determine the various processes involved in monomolecular photophysical decay including fluorescence and phosphorescence and give examples of photosensitised reactions.
7. Formulate quantum yield calculations in photochemistry.
8. Verify the relevance of photochemistry in a range of scientific fields including the pharmaceutical sector.

Learning and Teaching Methods

- Lectures.
- Additional knowledge of current research in area, independently sourced by student
- Set examples.
- Exercises.
- Problem sets.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5,6,7,8

Assessment Criteria

<40%: No understanding of the basic principles of inorganic and photochemistry.
40%-49%: Is able to understand the principles of inorganic and photochemistry. May have some difficulty with applying this theory.
50%-59%: All the above in addition to evaluation of photochemistry and catalytic cycles. Correctly able to assign point groups.
60%-69%: In addition to the above, shows a high level of competence and efficiency in the subject area.
70%-100%: All previous to excellent level. Be able to describe and discuss inorganic molecules and photochemical properties of complexes.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- Crabtree, R.H. _The organometallic Chemistry of the Transition Metals_. 5th ed. Signapore: Wiley Publishing, 2009.
- Fenton, D.E. _Biocoordination Chemistry_. 1st ed. UK: Oxford Science, 2008.

- Turro, N.J. _Modern Molecular Photochemistry_. USA: University Science Books, 1991.
- Wayne, C. and R. Wayne. _Photochemistry_. UK: Oxford Science, 1996.
- Willock, D. _Molecular Symmetry_. 1st ed. Signapore: John Wiley & Sons, 2009.

Supplementary Material

- Atkins, P.W. _Physical Chemistry_. 7th ed. UK: Oxford University Press, 2004.
- Cotton, F.A., G. Wilkinson and P.L. Gaus. _Basic Inorganic Chemistry_. 3rd ed. USA: John Wiley and Sons, 1995.
- Kettle, S.F.A. _Physical Inorganic Chemistry - A Coordination Chemistry Approach_. UK: Spektrum, 2013.